

11.2.2 CLOTHOID

The determination of the light intensity can be built geometrically using the Clothoid or Cornu's spiral. We show in this chapter two cases: (1) diffraction by an edge, and (2) diffraction by a rectangular aperture.

11.2.2.1 Clothoid Properties

By plotting $S(x)$ versus $C(x)$ we obtain the famous curve named "Clothoid" or Cornu's spiral. This operation is similar to plotting the function $F(x) = C(x) + iS(x)$ in the complex plane. The variable x corresponds to the curvilinear abscissa which is tagged in red on the curve.

As we can see, we obtain the behavior previously noticed in [Figure 11.27](#), namely the convergence to ± 0.5 when x increases. This convergence is demonstrated in a detailed manner in the plots of [Figure 11.30](#).

Let us now consider two points of the curve symmetrically positioned and draw a straight line connecting these two points ([Figure 11.31](#)). The length of this line segment is

$$P = [(C(x) - C(-x))^2 + (S(x) - S(-x))^2]. \quad (11.21)$$

This expression is similar, up to a multiplicative constant, to one of the two terms of Equation (11.18) which is nothing but the variation of the light intensity in one unique direction.

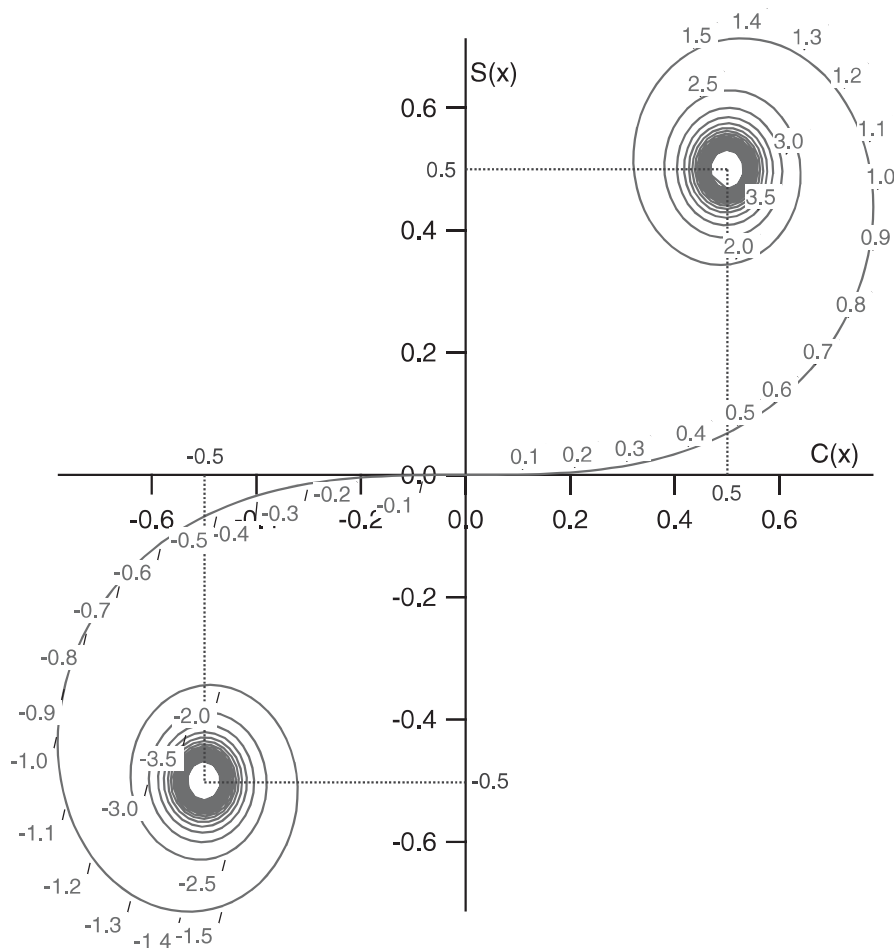


Figure 11.29: Representation of the Clothoid for $x \in [-10, 10]$.

```

1  #— Plot of the Spirale —
2  mplids=self.mplwidget_spiral.canvas
3  mplids.ax.clear()
4  mplids.ax.grid(True)
5  mplids.ax.spines['left'].set_position('center')
6  mplids.ax.spines['right'].set_color('none')
7  mplids.ax.spines['bottom'].set_position('center')
8  mplids.ax.spines['top'].set_color('none')
9  mplids.ax.spines['left'].set_smart_bounds(True)
10  mplids.ax.spines['bottom'].set_smart_bounds(True)
11  mplids.ax.xaxis.set_ticks_position('bottom')

```

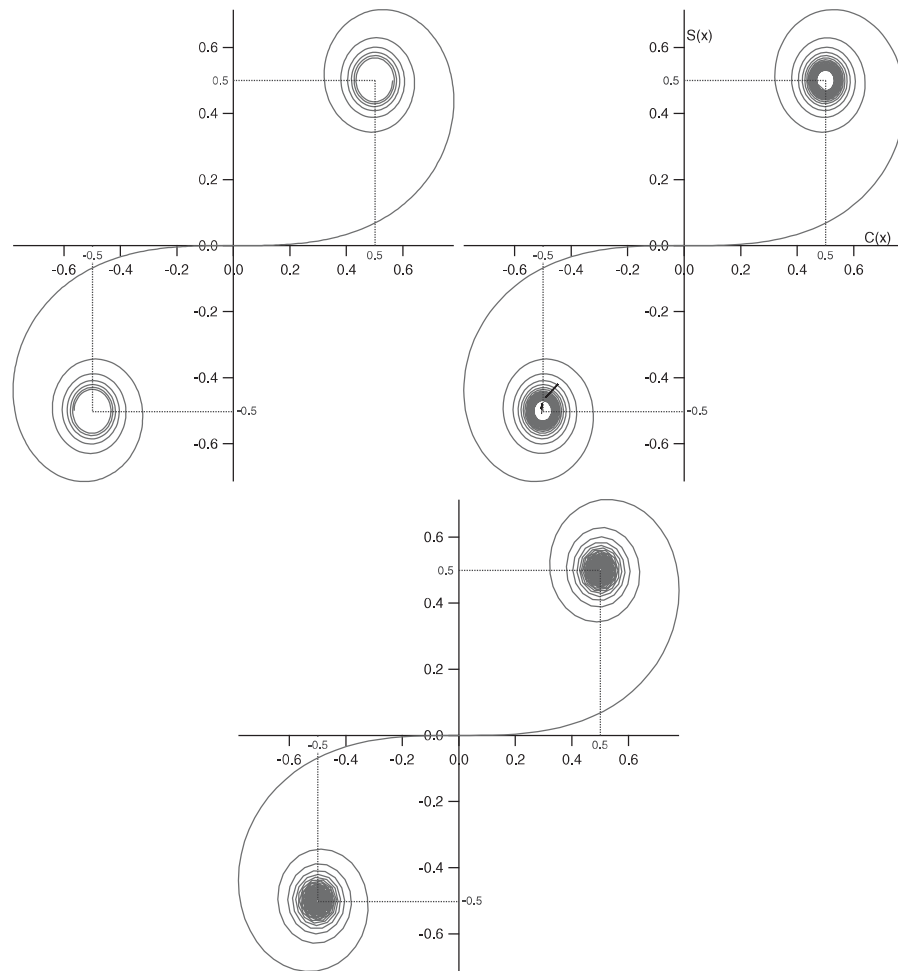


Figure 11.30: Clothoid for three intervals: Top left: $x \in [-5, 5]$, top right: $x \in [-10, 10]$, and bottom: $x \in [-20, 20]$.

```

12 mplids.ax.yaxis.set_ticks_position('left')
13 mplids.ax.plot(FC,FS,'k-',label='x=C,y=S',linewidth=2,alpha=0.8)
14 ind = arange(-2.,0,0.2)
15 s_c,s_s = self.Fresnel_C(ind,len(ind)),self.Fresnel_S(ind,len(ind))
16 for i in range(len(ind)):
17     mplids.ax.annotate("%.4s"%str(ind[i]),xy=(s_c[i],s_s[i]),
        xycoords='data', xytext=(0, -1),textcoords='offset points',
        arrowprops=dict(arrowstyle="—"),fontSize=8,color='red',
        bbox=dict(boxstyle="round", fc=(1,1,1), ec="none",alpha

```

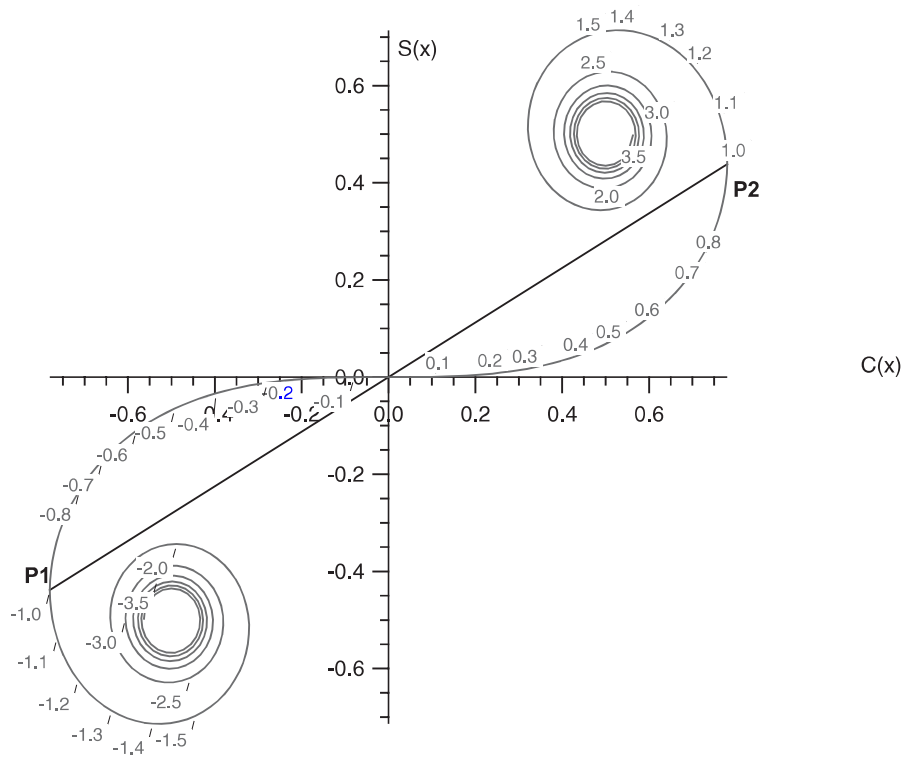


Figure 11.31: Representation of the Clothoid for $x \in [-10, 10]$.

```

        =0.5), horizontalalignment='right', verticalalignment='top'
    )
18 ind = arange(0,2.,0.2)
19 s_c,s_s = self.Fresnel_C(ind,len(ind)),self.Fresnel_S(ind,len(ind))
20 for i in range(len(ind)):
21     mpl1d.ax.annotate("%4s"%str(ind[i]),xy=(s_c[i],s_s[i]),
        xycoords='data',xytext=(0, -1),textcoords='offset points',
        arrowprops=dict(arrowstyle="—"),fontSize=8,color='red',bbox
        =dict(boxstyle="round", fc=(1,1,1), ec="none",alpha=0.8),
        horizontalalignment='right', verticalalignment='top')
22 mpl1d.draw()

```

Listing 11.8: Python script for the plot of the Clothoid, [Figures \(11.29\)](#) and [\(11.30\)](#).

11.2.2.2 Diffraction by a Single Slit

We now study the general case of rectangular aperture but this will not add significant meaning to the demonstration we showed using the Clothoid. To show this, let's consider the case for which h is very large ($h \gg 1$). The limits of integration are

$$v_1 = \frac{-h}{2} \sqrt{2 \frac{d+D}{\lambda d D}}$$

and

$$v_2 = \frac{h}{2} \sqrt{2 \frac{d+D}{\lambda d D}}.$$

Consequently, $C(v_1)$ and $C(v_2) \rightarrow 0.5$. Similarly, $S(v_1)$ & $S(v_2) \rightarrow 0.5$. Therefore, the second term in Equation (11.18) $\rightarrow 2$, and we obtain once again Equation (11.21). Thus, we can continue the study of diffraction by a slit without loss of generality.

We will now see how to calculate the light intensity along the x direction of the observation screen. Before doing this, we have to define precisely the integration limits on the Fresnel integrals. We will use the notations of [Figure 11.25](#) for a slit of width b . These limits, named (x_1) and (x_2) , depend on the observation point x_M and more precisely on the position of the origin in the slit. This is defined as the intersection between the axis joining the source of light to x_M and the slit's plane (D) as shown in [Figure 11.32](#).

Hence, when x_M is on the z -axis the integration limits are $x_1 = -b/2$ and $x_2 = b/2$. These change when we move x_M as shown in [Figure 11.32](#). In this case, we have $x_1 = x - (b/2)$ and $x_2 = x + (b/2)$. Given the fact that $x = x_M d / (d + D)$, we have

$$x_2 = \frac{d}{d+D} x_M + \frac{b}{2}$$

$$x_1 = \frac{d}{d+D} x_M - \frac{b}{2}.$$

Using these in $u = x \sqrt{2(d+D)/(\lambda d D)}$, we obtain

$$u_2 = \left(\frac{dx_M}{d+D} + \frac{b}{2} \right) \sqrt{2 \frac{d+D}{\lambda d D}}$$

$$= \sqrt{\frac{2d}{D(d+D)\lambda}} x_M + \frac{w}{2} \tag{11.22}$$

and

$$u_1 = \left(\frac{dx_M}{d+D} - \frac{b}{2} \right) \sqrt{2 \frac{d+D}{\lambda d D}}$$

$$= \sqrt{\frac{2d}{D(d+D)\lambda}} x_M - \frac{w}{2} \tag{11.23}$$

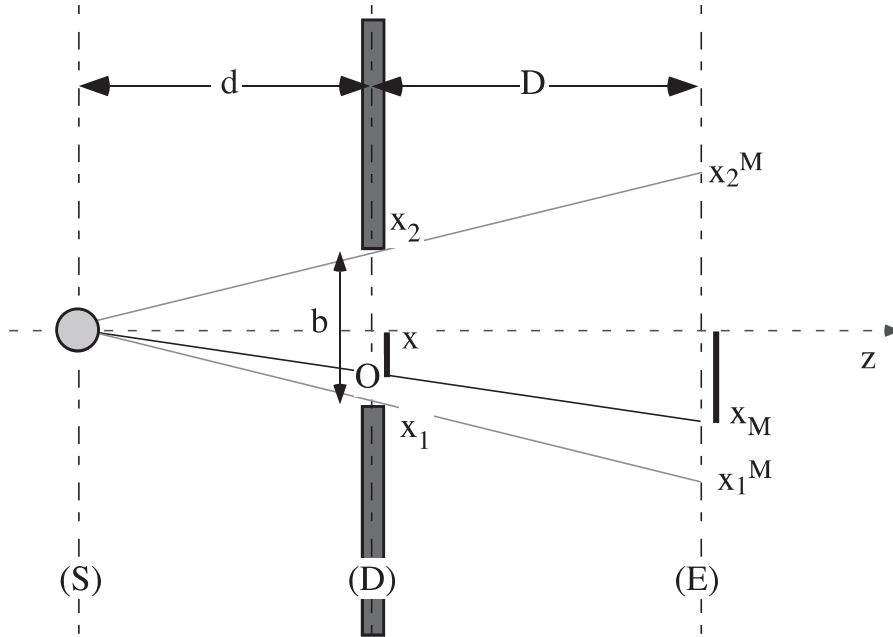


Figure 11.32: Schema of diffraction by a single slit with appropriate notations.

where we have

$$w = \sqrt{2 \frac{(d+D)}{\lambda d D}} (x_2 - x_1) = \sqrt{2 \frac{(d+D)}{\lambda d D}} b. \quad (11.24)$$

Therefore, w which is equal to the distance between the curvilinear abscissa of the Clothoid is directly related to the width of the slit.

Let us consider an example with $d = D = 1 \text{ m}$, $\lambda = 592 \text{ nm}$, $u_1 = -0.75(P_1)$, and $u_2 = 0.75(P_2)$ as shown in Figure 11.33. It is easy to see that $w = u_2 - u_1 = 1.5$, so $b = w/2600 = 0.58 \text{ mm}$. The illumination on the observation screen on the axis of symmetry z (namely $x_M = u = 0$) is equal to the modulus of the segment (also called rope) $[P_1, P_2]$ times $I_0/2$. The intensity is then

$$\begin{aligned} I &= \frac{I_0}{2} |P_2 P_1|^2 \\ &= \frac{I_0}{2} \left\{ [S(0.75) - S(-0.75)]^2 + [C(0.75) - C(-0.75)]^2 \right\} \\ &= \frac{I_0}{2} [(0.21 + 0.21)^2 + (0.7 + 0.7)^2] \\ &= 1.07 I_0. \end{aligned}$$

So, for $I_0 = 1$, $I = 1.07$. Therefore, the light intensity in the central point

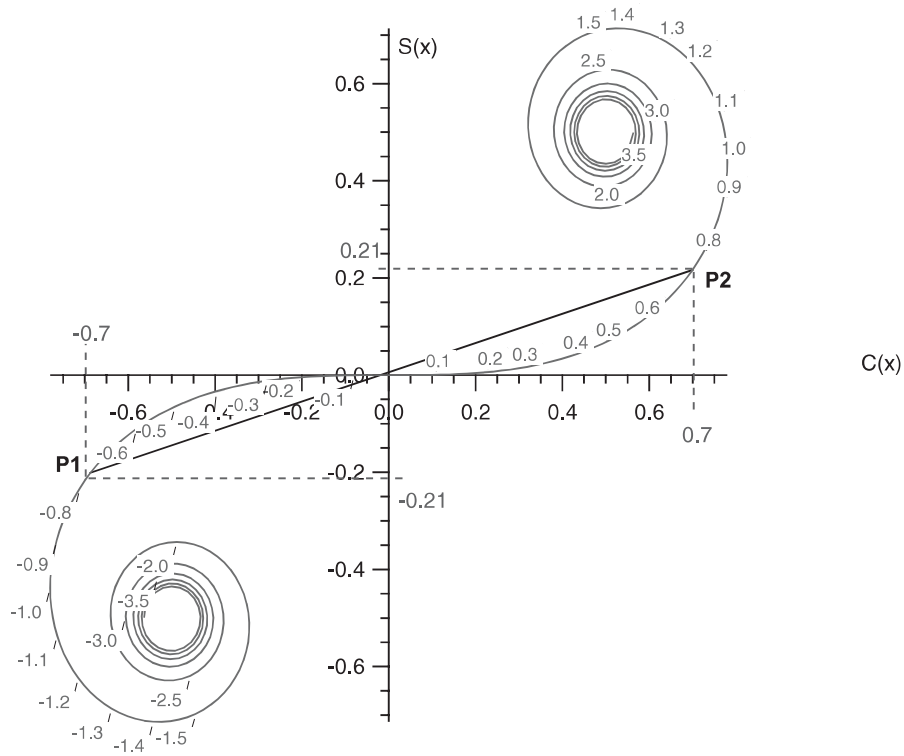


Figure 11.33: Cornu's spiral for slit of width $b = 0.58 \text{ mm}$ ($w = 1.5$).

of the observation screen for a slit of width $b = 0.58 \text{ mm}$ and $I_0 = 1$ is equal to $I = 1.07$. If we want to know the distribution of the light intensity along the x -axis we have to move the two points (P_1, P_2) along the spiral from one side or the other. The symmetry of the problem will lead to the same behavior for both sides. Let's move the rope $[P_1, P_2]$ to the right side. For each displacement we should record the position of the middle of the rope $(P_2 + P_1)/2$ and its modulus.

This operation could be tricky especially when we want to store a large number of points for the curve. For this reason it's interesting to develop a program with an interface that allows us to do this operation easily. [Figure 11.34](#) is a screen capture of this interface.

```

1 P1=P1_old + Delta_PushButton
2 P2=P2_old + Delta_PushButton
3 dP1=P1*1.001
4 dP2=P2*1.001
5 u1=linspace(P1,dP1,1)

```

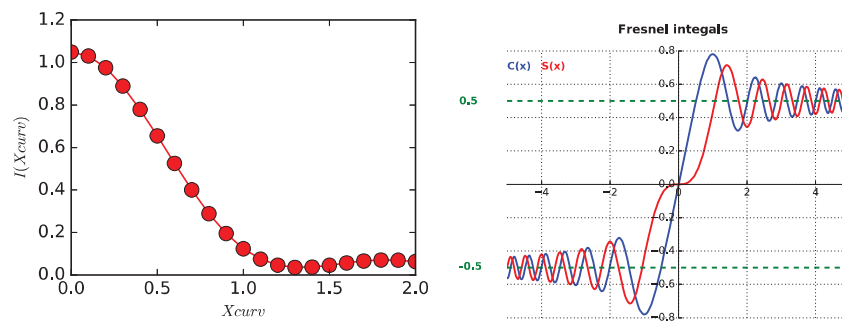
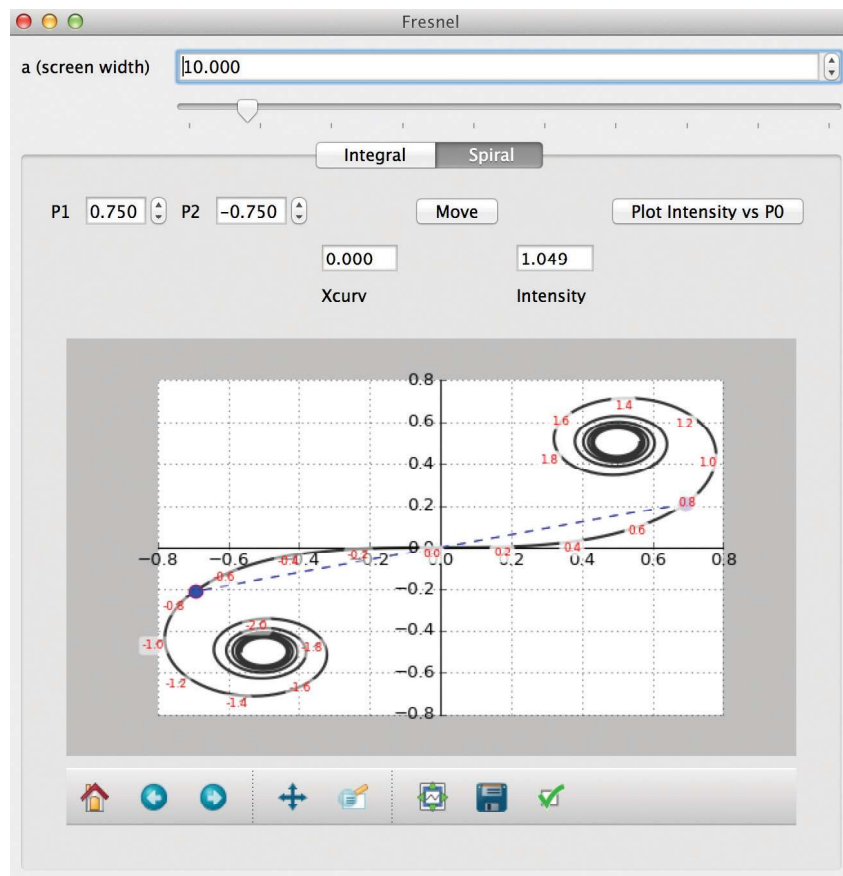


Figure 11.34: Program used to generate light intensity from the Clothoid.


```

6  u2=linspace(P2,dP2,1)
7  FCp1,FSp1=self.Fresnel_C(u1,1),self.Fresnel_S(u1,1)
8  FCp2,FSp2=self.Fresnel_C(u2,1),self.Fresnel_S(u2,1)
9  Intensity = sqrt((FCp1[0]-FCp2[0])**2 + (FSp1[0]-FSp2[0])**2)
10 Intensity = .5*Intensity*Intensity
11 X_Intensity = (P1+P2)/2
12 self.lineEdit_XIntensity.setText("%.3f" %X_Intensity)
13 self.lineEdit_Intensity.setText("%.3f" %Intensity)
14 self.f=open('Intensity.txt','a')
15 self.f.write(str(X_Intensity)+' '+str(Intensity)+'\n')
16 self.f.close()
17 mpllids.ax.plot([FCp1,FCp2], [FSp1,FSp2],linestyle='—', marker='o',
18                 markersize=10,markeredgecolor='r')
19 mpllids.draw()
20 P1_old=P1
21 P2_old=P2

```

Listing 11.9: Python script calculating the shifting of the segment P_1P_2 in Figures 11.33 and 11.34.

```

1  x,I=loadtxt("Intensity.txt", delimiter=",", unpack=True)
2  if plt.figure_exists(1):
3      plt.close()
4  fig=plt.figure(figsize=(5,3))
5  fig.canvas.manager.window.move(800,20)
6  fig.canvas.set_window_title('Intensity')
7  plt.plot(x,I,linestyle='—',marker='o',color='r',markersize=10,
8          markeredgecolor='b')
9  plt.ylim(0,1.6)
10 plt.title('Fresnel diffraction', fontsize=16, weight='bold')
11 plt.xlabel('x', fontsize=12, weight='bold')
12 plt.ylabel('I (x)', fontsize=12, weight='bold')
13 plt.grid()
14 plt.show()

```

Listing 11.10: Python script for the Intensity versus Xcurv graph in Figure 11.34. This script shows how the intensity curve is continuously built each time we press the “Move” push-button.

We show in Figure 11.35 five records chosen arbitrarily from a large number of records used to plot the light intensity in the middle-left panel. To complete the light intensity along the x -axis (for the negative values of x), we just duplicate and flip the resultant distribution as shown in Figure 11.36. Finally,

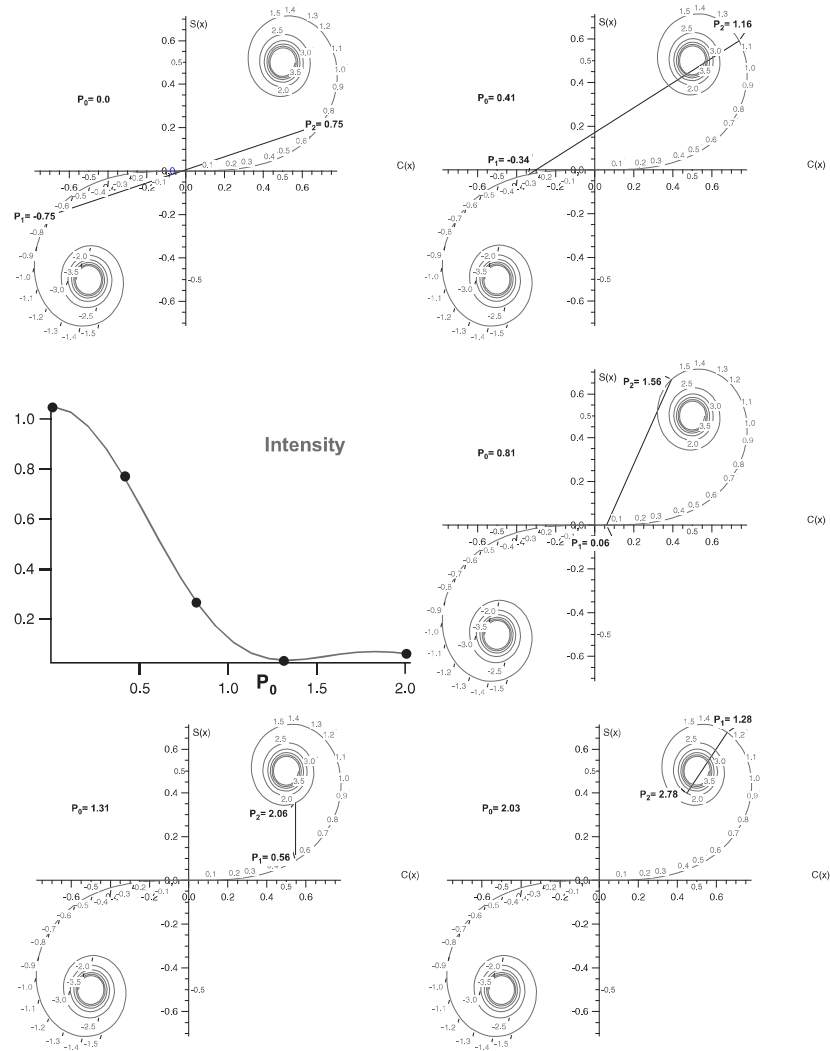


Figure 11.35: Figure showing the procedure used to draw the light intensity (middle-left panel).

we can clearly show the behavior of the sinc function obtained previously in [Section 11.1](#).

Let us now consider the case of single slits of arbitrary widths. The plots shown in [Figure 11.37](#) correspond to $w = 1.5, 2.5, 3.9, 4.6, 5.2, 6.2, 12$. Therefore, for configurations similar to the previous one ($d = D = 1m$ and $\lambda = 592nm$) these curves describe the diffraction for slits of width

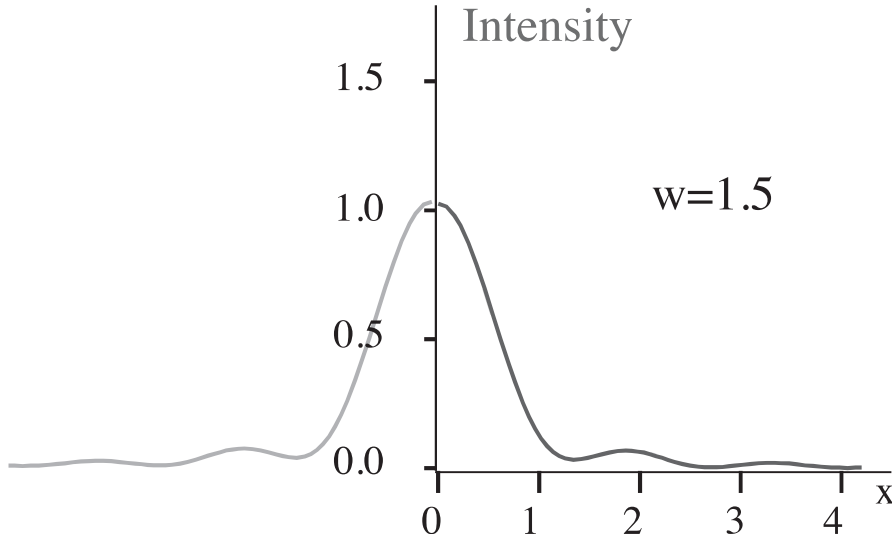


Figure 11.36: Light intensity obtained from diffraction by slit with $b = 0.58 \text{ mm}$.

$b = 0.58 \text{ mm}$, 0.96 mm , 1.5 mm , 1.8 mm , 2 mm , 2.4 mm , and 4.6 mm , respectively.

It is worth noting here the differences between the first two curves ($w = 1.5$, and $w = 2.5$). In the first case, the curve corresponds to the Fraunhofer diffraction while the second clearly departs from it. We can check from here the Fresnel criteria ($b^2/(4\lambda D) \ll 1$). For $w = 1.5$ ($b = 0.58 \text{ mm}$), $b^2/(4\lambda D) = 0.14 \ll 1$ and for $w = 2.5$ ($b = 0.96 \text{ mm}$), $b^2/(4\lambda D) = 0.40 < 1$.

This confirms what we have observed on the two curves, that is, we pass from the Fraunhofer to Fresnel diffraction regime when we increase b from 0.58 mm to 0.96 mm . The slit has to be slightly enlarged to be clearly in the Fresnel regime and for the third case we set $w = 3.9$, $N_F \simeq 1$.

11.2.2.3 Diffraction by an Edge

We take again the same notations as previously with d the distance between the source plane and the edge plan, D the distance between the observation screen and the edge plan. We consider the configuration used in the previous section, namely $d = D = 1 \text{ m}$, and $\lambda = 592 \text{ nm}$. We also fix $I_0 = 1$.

In the case of diffraction by an edge, the plane of the edge reduces to the half-plane. One of the two edges is sent to infinity. The limits for the Fresnel integrals are obtained from the values of x which varies from $0 \rightarrow \infty$. But the origin for the variable x is defined by the intersection of the line joining the source to the observation point with the plane of the diaphragm. Therefore, we have to consider separately the part of the observation screen hidden by the obstacle, namely, the area of the geometrical shadow, and the directly lit part.

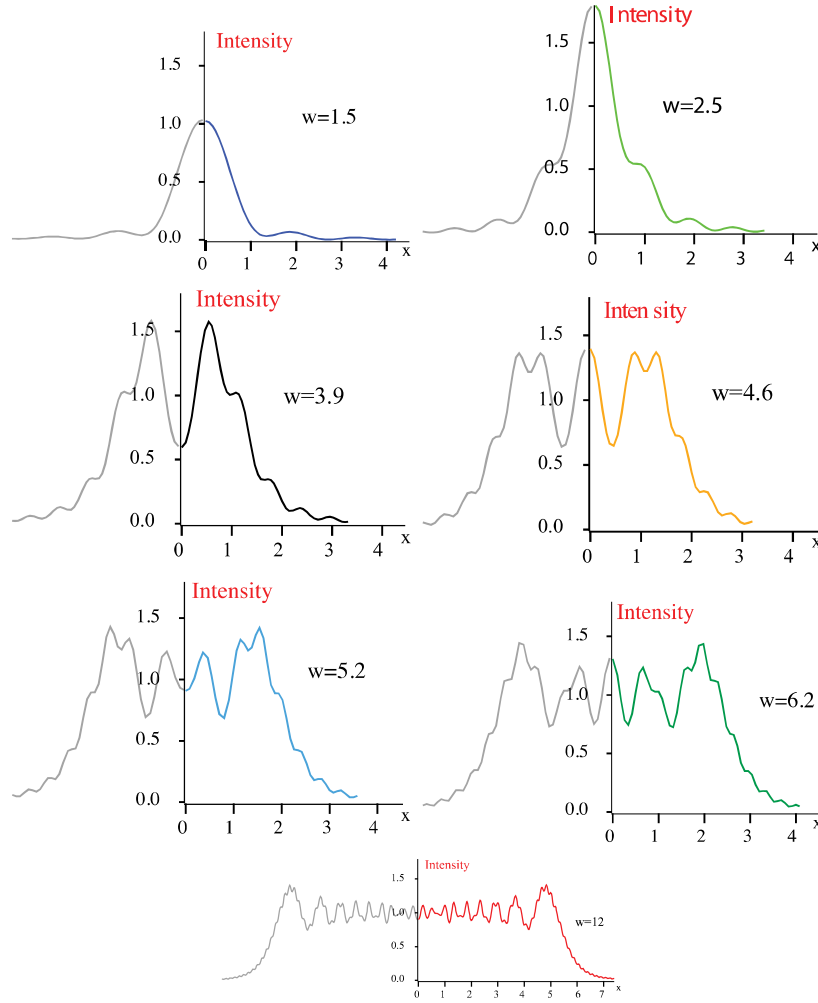


Figure 11.37: Diffracted intensity for various slits obtained by the Clothoid.

In both cases, we express the relation between x and x_M from geometrical considerations.

$$\begin{aligned}
 x &= \frac{d}{d+D} x_M \\
 \Rightarrow u &= \sqrt{2 \frac{d+D}{dD\lambda}} x \\
 &= \sqrt{2 \frac{d}{D(d+D)\lambda}} x_M.
 \end{aligned} \tag{11.25}$$

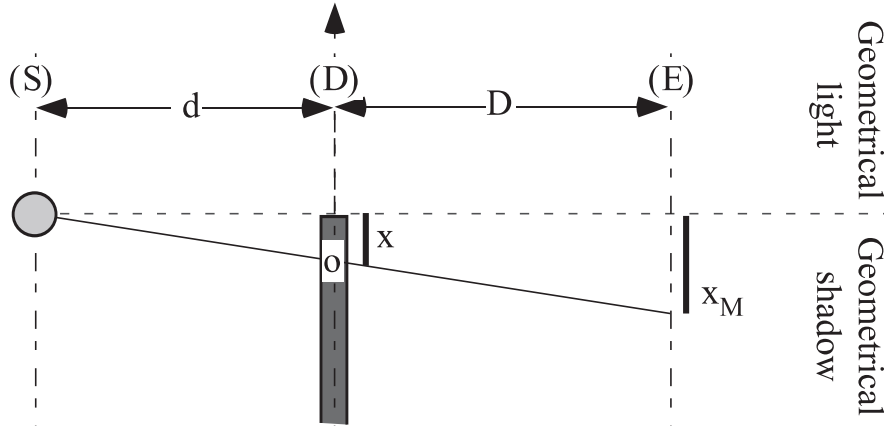


Figure 11.38: Schema of the edge with the adopted notations.

We recall here that u is the curvilinear abscissa of the Clothoid and that it corresponds to the limits of integration for the Fresnel integrals.

- *Geometrical shadow*

When we go forward into the geometric shadow, the origin for the variable x in the edge plane goes down simultaneously. Therefore, the edge moves away from the origin (O) in the $x > 0$ direction as shown in Figure 11.38. In this case, the limits of integration are $x \rightarrow +\infty$ on the visible side and the other side remains at $+\infty$.

Therefore, we only use the positive part of the Clothoid. Of course, if we change the convention of the sign in Figure 11.38, we will have to keep the negative part of the Clothoid. To get the intensity in an arbitrary position of x_M we simply need to measure the modulus of the rope $|P_x P_\infty|$. To simplify, we also set $I_0 = 1$ as previously. To illustrate this by an example, let us choose $u = 0.7$ on the Clothoid as is shown in Figure 11.39.

The illumination at the point $x_M = 0.7/1300 = 0.54 \text{ mm}$ [Equation (11.25)] is given by

$$\begin{aligned}
 I &= \frac{1}{2} |P_x P_\infty|^2 \\
 &= \frac{1}{2} \left\{ [S(\infty) - S(0.7)]^2 + [C(\infty) - C(0.7)]^2 \right\} \\
 &= \frac{1}{2} [(0.5 - 0.18)^2 + (0.5 - 0.66)^2] \\
 &= 0.064.
 \end{aligned}$$

At the border between geometrical shadow and geometrical light ($x_M = 0$), x is the point P_0 on the Clothoid. The modulus of $|P_0 P_\infty|$ is simply $1/2$, and we obtain the well-known result, namely,

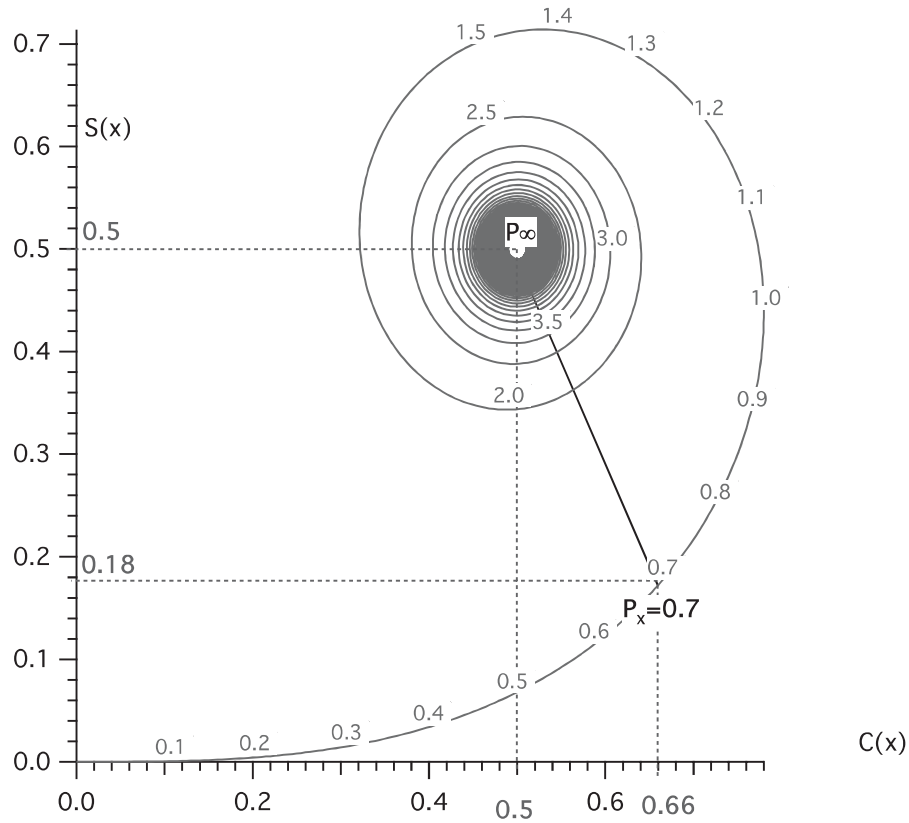


Figure 11.39: Light intensity obtained from diffraction by slit with $b = 0.58 \text{ mm}$

$$I = \frac{1}{2} \left[\frac{1^2}{2} + \frac{1^2}{2} \right]$$

$$\Rightarrow I = 0.25$$

As shown previously, we can draw the variations of light intensity in the shadow part by moving the point P_0 along the Clothoid.

- *Geometrical light*

We need the double spiral to treat the lit area. Indeed, when we go along the illuminated part, as shown in [Figure 11.41](#), the origin increases simultaneously to x_M . Therefore the edge goes away from the origin on the negative side. The limits are now $x \rightarrow -\infty$ in the visible side and the other side remains at $+\infty$.

Let us consider an example with $u = 0.8 \Rightarrow x_M = 0.62 \text{ mm}$ on the illuminated part of the observation screen.

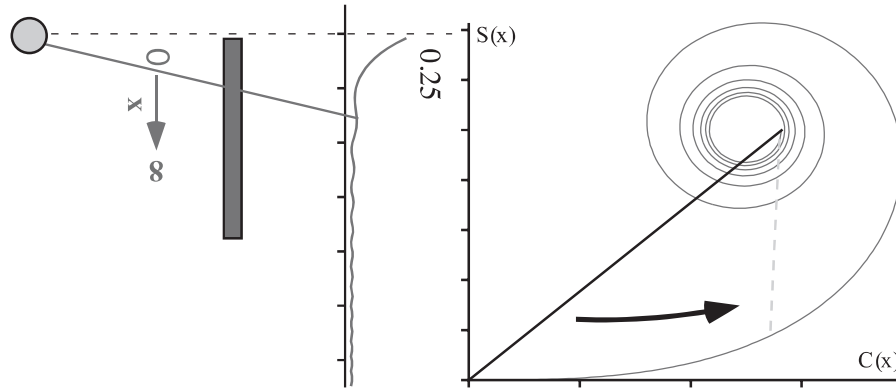


Figure 11.40: Distribution of the light intensity in the geometrical shadow (left) and the schema showing how to proceed with the Clothoid (right).

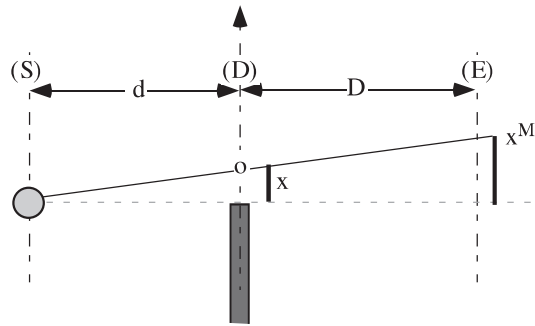


Figure 11.41: Schema for the lighted part with its notations.

$$\begin{aligned}
 I &= \frac{1}{2} |P_x P_\infty|^2 \\
 &= \frac{1}{2} \left\{ [S(\infty) - S(-0.8)]^2 + [C(\infty) - C(-0.8)]^2 \right\} \\
 &= \frac{1}{2} [(0.5 + 0.72)^2 + (0.5 - 0.25)^2] \\
 &= 1.02.
 \end{aligned}$$

Figure 11.43 summarizes the complete distribution of the light intensity for both geometrical shadow and geometrical light.

11.2.3 DIFFRACTION BY A SINGLE SLIT

We showed in the previous section how to calculate the diffraction of the light by the use of the Clothoid. We now show how to calculate it directly from

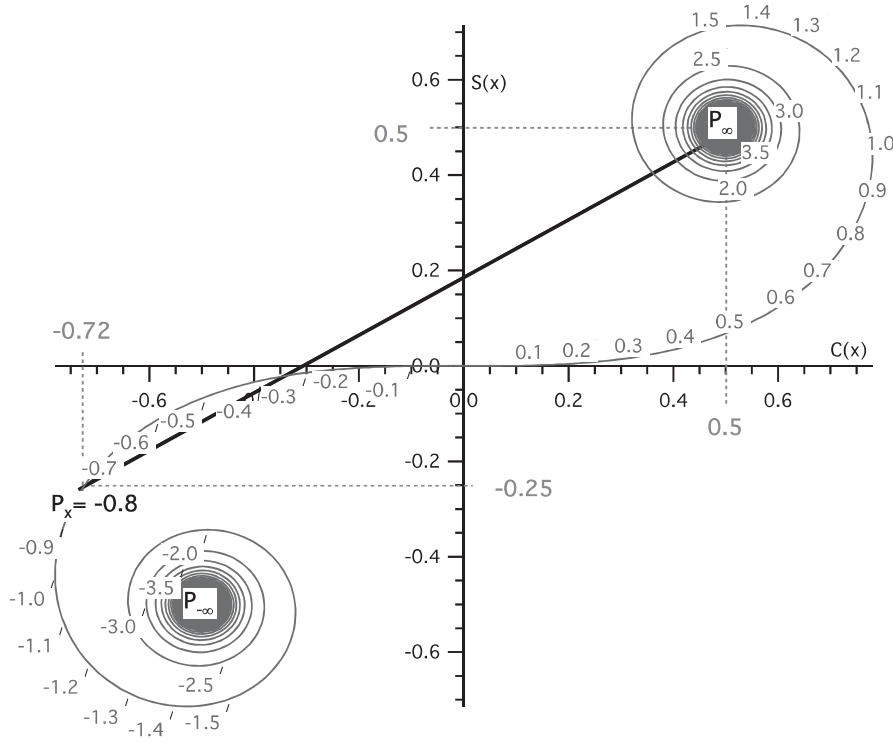


Figure 11.42: Calculation of the intensity in the lit part.

Equation (11.18). We continue with the same expressions introduced earlier [Equations (11.22), (11.23), and (11.24)].

$$u_2 = \sqrt{\frac{2d}{D(d+D)\lambda}} x_M + \frac{w}{2}$$

$$u_1 = \sqrt{\frac{2d}{D(d+D)\lambda}} x_M - \frac{w}{2}$$

and $w = b\sqrt{2\frac{(d+D)}{\lambda d D}}.$

When $d \rightarrow \infty$, $w \rightarrow b\sqrt{2/(\lambda D)}$. Therefore,

$$u_2 = \sqrt{\frac{1}{2D\lambda}} (2x_M + b)$$

$$u_1 = \sqrt{\frac{1}{2D\lambda}} (2x_M - b).$$

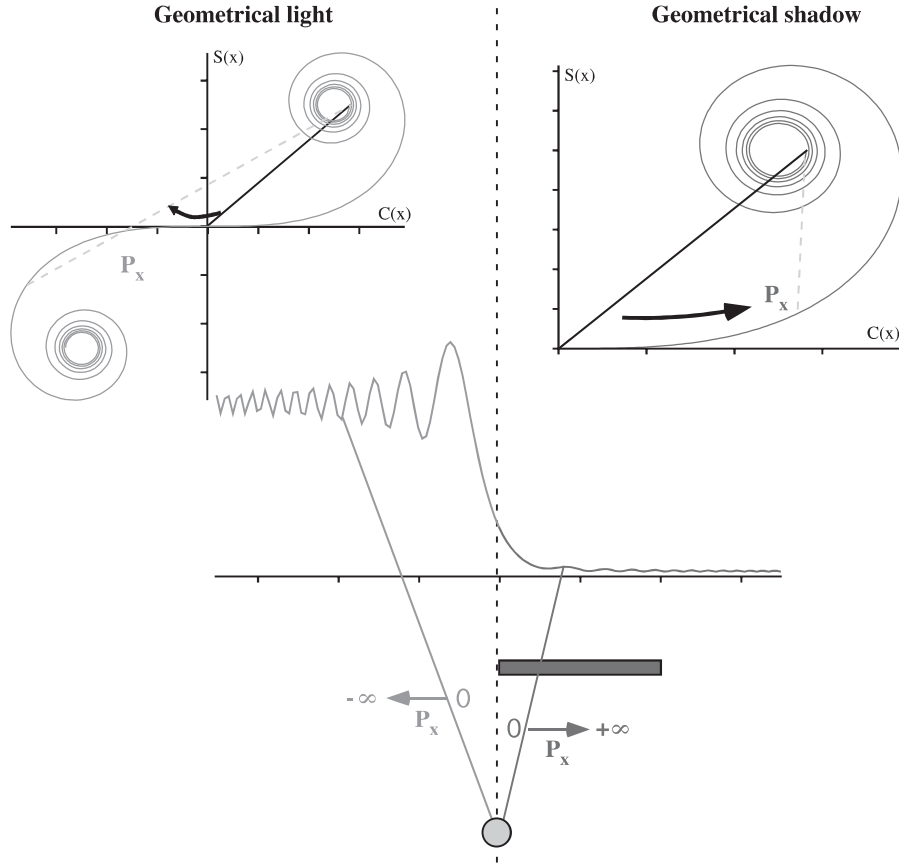


Figure 11.43: Complete process to calculate the diffraction by an edge.

We do the same for the y -direction:

$$v_2 = \sqrt{\frac{1}{2D\lambda}}(2y_M + h)$$

$$v_1 = \sqrt{\frac{1}{2D\lambda}}(2y_M - h).$$

The light intensity in each direction is then

$$I_x = \frac{1}{2} \left\{ [C(u_2) - C(u_1)]^2 + [S(u_2) - S(u_1)]^2 \right\}$$

$$I_y = \frac{1}{2} \left\{ [C(v_2) - C(v_1)]^2 + [S(v_2) - S(v_1)]^2 \right\}. \quad (11.26)$$

We show as an example the behavior of these two functions for an aperture of arbitrary dimension; here as width $b = 2 \text{ mm}$ and height $h = 0.5 \text{ mm}$. The

wavelength is $\lambda = 592 \text{ nm}$ and the observation screen is at a distance $D = 1 \text{ m}$.

We can clearly see in Figure 11.44 that the diffraction by the width is of Fresnel type and the one produced by the height is of Fraunhofer type. We can check this by verifying the Fresnel criteria. For the width, $b^2/(4\lambda D) = 1.67$ and for the height $h^2/(4\lambda D) = 0.1$.

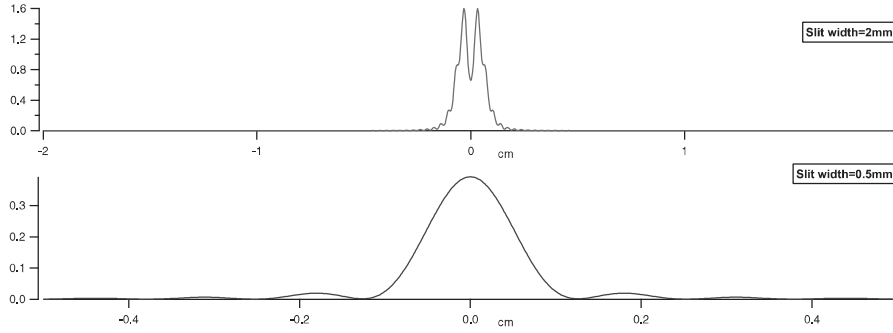


Figure 11.44: Diffraction by rectangular aperture of $b = 2 \text{ mm}$ and $h = 0.5 \text{ mm}$. The top curve represents the intensity in the x -direction and the bottom one the intensity in the y -direction

The product of the two functions in Equation (11.26), which is nothing but Equation (11.18), leads to the diffraction pattern in two dimensions as illustrated for different cases in Figure 11.45. The last case (bottom-right image) corresponds to aperture of both large width and height.

We notice, from the examples, that the light intensity shows variations passing through a succession of maxima and minima. It is possible to predict the number of these extrema by the use of the Fresnel number $N_F = b^2/(4\lambda D)$. Simply, the number of maxima is equal to N_F and the number of minima is equal to $N_F - 1$. The four examples in Figure 11.46 clearly show this property. We use the same values of $\lambda = 592 \text{ nm}$ and $D = 1 \text{ m}$ as in the previous simulations.

For instance, let's look at the top-right curve for which $b = 4.072 \text{ mm}$; so the Fresnel's number is $N_F = b^2/(4\lambda D) = 7$. Thus, as we can see on this curve, the number of maxima is 7 and the number of minima is 6.

11.2.4 DIFFRACTION BY AN EDGE

We start again with the expressions obtained for the treatment of the edge with the Clothoid. We assume that the source is at $d \rightarrow \infty$. Hence, Equation (11.25) becomes

$$u = \sqrt{2 \frac{d}{D(d+D)\lambda}} x_M = \sqrt{\frac{2}{D\lambda}} x_M.$$

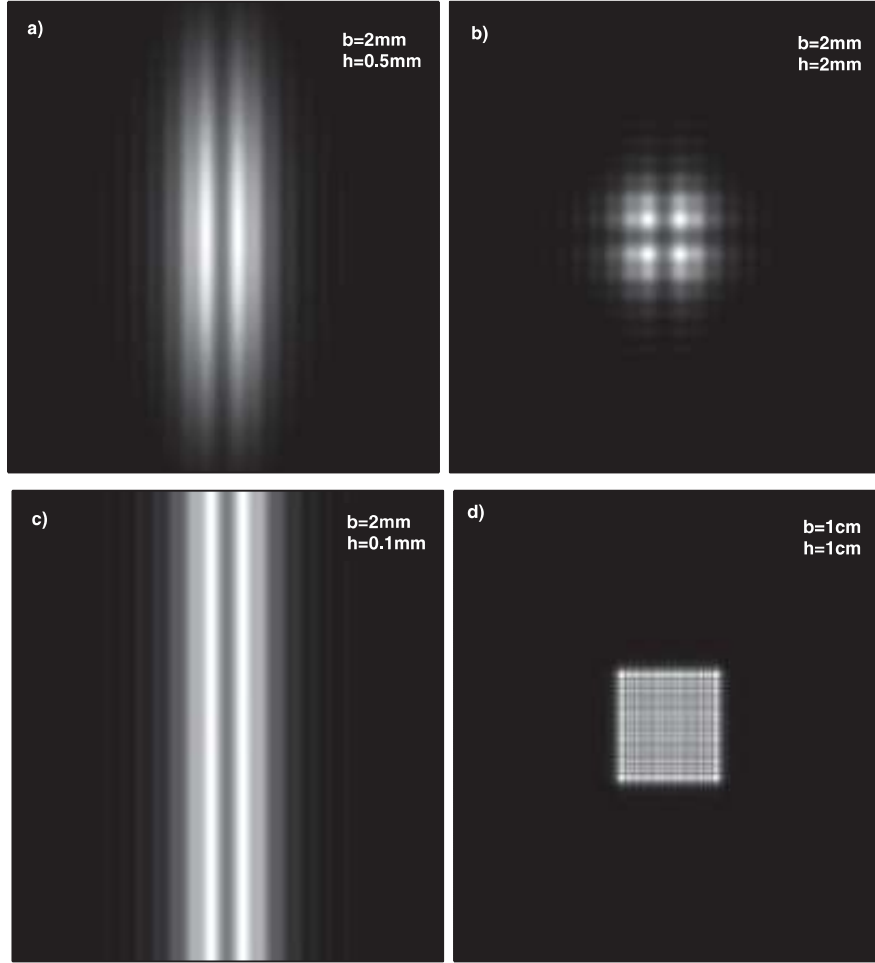


Figure 11.45: Images of diffraction for different sizes of the slits. The top-left image corresponds to the case of [Figure 11.44](#).

An edge may be considered as the boundary of a very wide slit. Let the height of the aperture $h \rightarrow \infty$. This means that both $v_1 \rightarrow -\infty$ and $v_2 \rightarrow \infty$ and that one of u_1 or $u_2 \rightarrow \infty$. Let's choose $u_2 \rightarrow \infty$ and $u_1 = u$. Equation (11.18) becomes

$$\begin{aligned}
 I &= \frac{1}{4} \left\{ [C(\infty) - C(u)]^2 + [S(\infty) - S(u)]^2 \right\} \\
 &\quad \left\{ [C(\infty) - C(-\infty)]^2 + [S(\infty) - S(-\infty)]^2 \right\} \\
 &= \frac{1}{2} \left\{ [0.5 - C(u)]^2 + [0.5 - S(u)]^2 \right\}.
 \end{aligned}$$

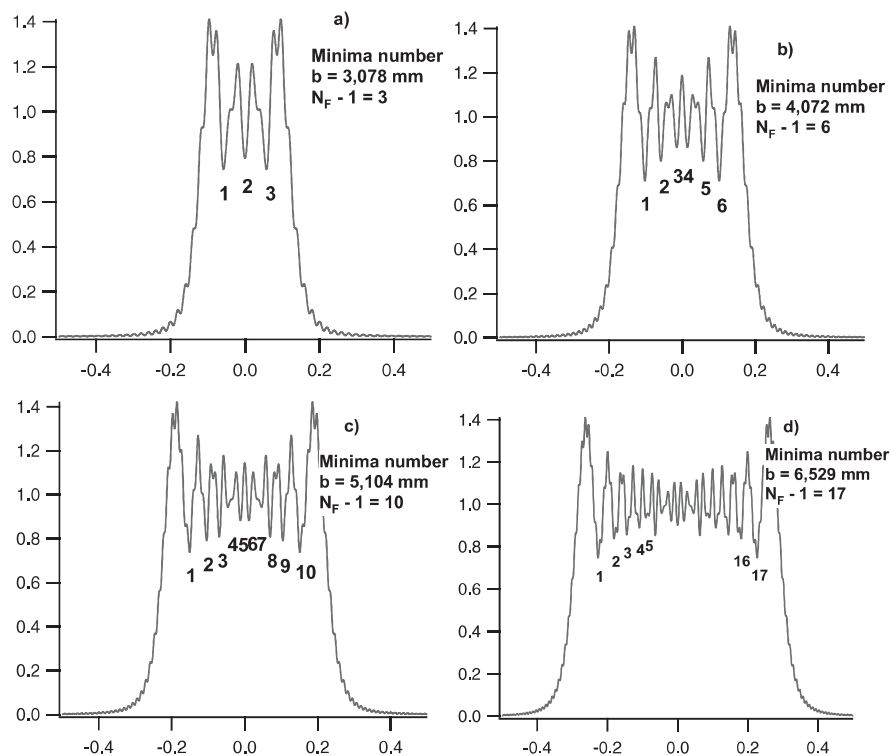


Figure 11.46: Verification of the diffraction pattern by the use of Fresnel's number.

Figure 11.47 shows the diffraction pattern for a wavelength $\lambda = 592 \text{ nm}$ and the observation screen at a distance $D = 1 \text{ m}$.

```

1  #— Input parameters —
2  Lambda=self.Slider_Lambda.value()*1.E-09
3  D=self.Slider_D.value()*1.E-02
4  N=500
5  X_Mmin= -5*1.E-3
6  X_Mmax= 5*1.E-3
7  coef = -sqrt(2/Lambda/D)
8  x=linspace(X_Mmin, X_Mmax,N)
9  x1 = coef*x
10 FC=self.Fresnel_C(x1,N)
11 FS=self.Fresnel_S(x1, N)
12
13 I=[]

```

Diffraction

```

14 for i in range(N):
15     ix = complex(0.5-FC[i],0.5-FS[i])
16     Intens=(real(ix)**2 + imag(ix)**2)/2.
17     I.append(Intens)
18 I2D_0=[]
19 I2D_1=[]
20 for i in range(N):
21     for j in range(2):
22         I2d=I[i]
23         I2D_0.append(I2d)
24     I2D_1.append(I2D_0)

```

Listing 11.11: Python script with the input parameters for the above intensity expression.

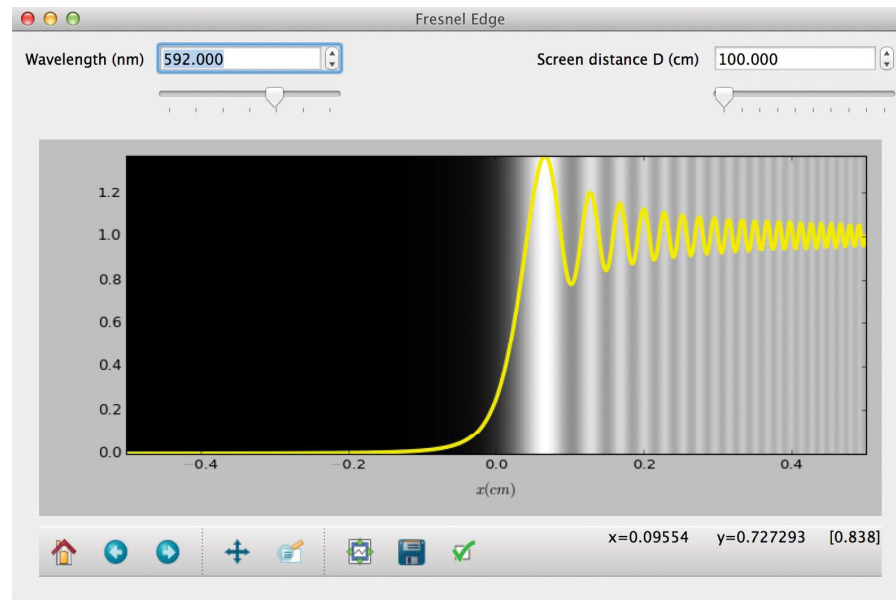


Figure 11.47: Complete process to calculate the diffraction by an edge.

```

1 #— Graphs —
2 mpl1d=self.mplwidget.canvas
3 mpl1d.ax.clear()
4 mpl1d.ax.plot(x*1E2,I,'-',color="#ffff00", linewidth=3)
5 mpl1d.ax.imshow(I2D_1,cmap=cm.gray,interpolation='bilinear', aspect
   = 'auto',origin='lower',extent=[min(x*1E2),max(x*1E2),min(I),max
   (I)])

```

```

6 mpld.ax.set_xlabel(r'$x$ (cm)$', fontsize=12, fontweight='bold')
7 mpld.figure.tight_layout(rect=[0.05,0,1,1])
8 mpld.draw()

```

Listing 11.12: Python script for the plot of [Figure 11.47](#).

It is interesting to compare here the diffraction obtained by a slit and the diffraction obtained by two close edges. To do this, let us consider the example of two edges that we bring closer. Each edge gives its own diffraction pattern represented by the two curves in the top of [Figure 11.48](#). We simply superimpose these two curves in the bottom curve.

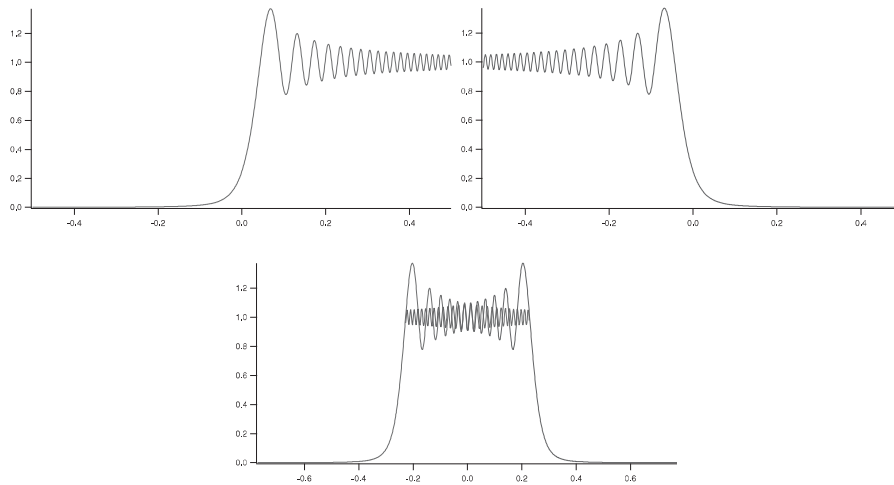


Figure 11.48: Diffraction by edges. The curves on the top give the diffraction by each edge and the bottom is a simple superimposition of the two.

The combination of these two curves consists of averaging the light intensity associated with the two edges. [Figure 11.49](#) shows the diffraction obtained by a slit of width $b = 5.26 \text{ mm}$ (left curve) and the diffraction obtained by two edges spaced by a distance 5.44 mm (right curve). Doing so, we don't take into account the interference between these two intensities. To do that we have to sum the amplitude instead of the intensity.

We repeat the same simulation for slit of width $b = 2.5 \text{ mm}$ in [Figure 11.50](#). One can notice again the good agreement between the two configurations despite the proximity of the two edges.

From these simulations, one can make the approximation that the diffraction by a slit is similar to the diffraction by two edges.

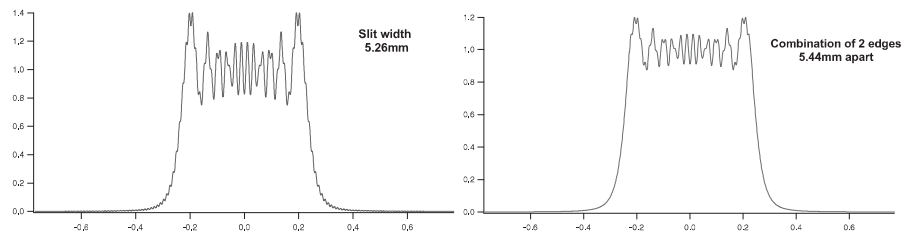


Figure 11.49: Diffraction by slit of width $b = 5.26 \text{ mm}$ (left) and two edges spaced by 5.44 mm (right).

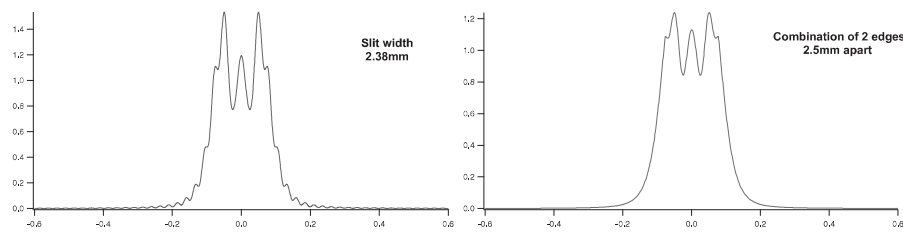


Figure 11.50: Diffraction by slit of width $b = 2.38 \text{ mm}$ (left) and two edges spaced by 2.5 mm (right).

REFERENCE

1. Vasudevan Lakshminarayanan and L Srinivasa Varadharajan. *Special Functions for Optical Science and Engineering*. SPIE, Bellingham, MA, 2016.